OXFORD

INTERNATIONAL AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

OXFORD AQA INTERNATIONAL A-LEVEL

BIOLOGY (9610)

Unit 2: Biological systems and disease

Wednesday 17 May 2017 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 75.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

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Answer **all** questions in the spaces provided.

0 1 A student prepares material from an onion root tip.

The student:

- removes 1 mm from the tip of the root
- places the root tip on a glass slide
- adds a drop of stain
- covers the material on the slide with a coverslip
- pushes down firmly on the coverslip, making sure the coverslip does not move sideways.

0 1 . 1 In the final step, the student pushes down firmly on the coverslip and makes sure the coverslip does **not** move sideways.

Explain why.

[2 marks]

0 1 . 2 When the student used a light microscope to observe the stained cells, the image could not be seen clearly.

Suggest **two** ways the student could adjust the microscope to make the image clearer.

[2 marks]

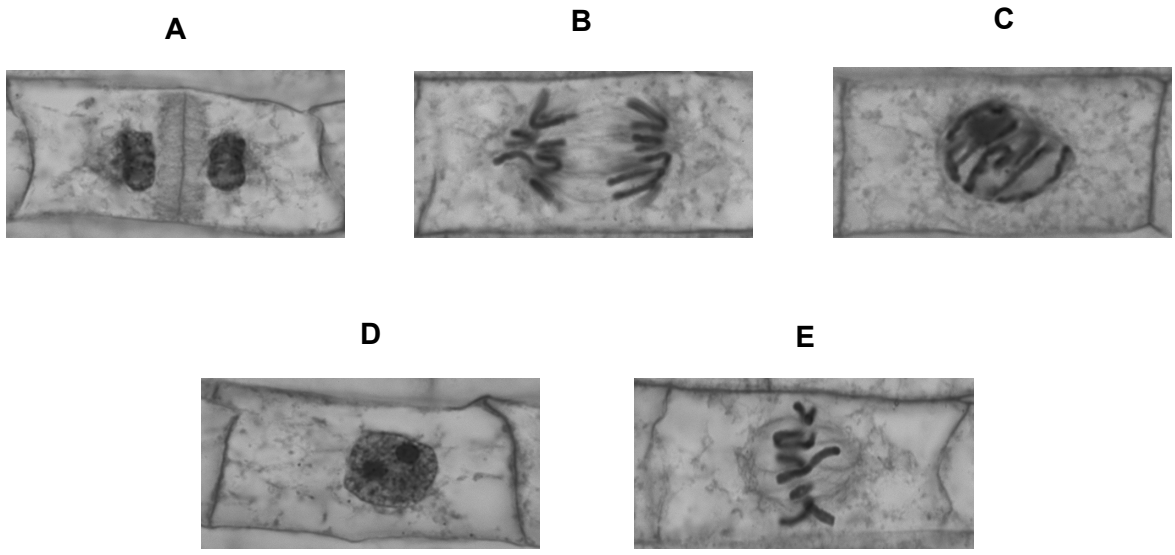
1 _____

2 _____

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Figure 1 shows cells from the onion root tip at different stages of the cell cycle.

Figure 1



0 1 . 3 Give the letters of stages **A** to **E** to show the correct order of the cell cycle.
Start with stage **D**.

[1 mark]

D → ☐ → ☐ → ☐ → ☐

0 1 . 4 Give the letter of the stage that shows a cell in anaphase.
Give **one** reason for your choice.

[2 marks]

☐

Reason _____

_____

Question 1 continues on the next page

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The student counted the number of cells observed in each stage of the cell cycle. **Table 1** shows the results.

Table 1

Stage of the cell cycle	Number of cells observed
Interphase	43
Prophase	21
Metaphase	8
Anaphase	5
Telophase	3

0 1 . 5 Calculate the mitotic index for this sample of cells.
Use data from **Table 1**.

[2 marks]

Mitotic index = _____

0 1 . 6 Assume that the cell cycle for cells in onion root tips takes 12 hours.
Calculate the time, in minutes, during which these cells are in metaphase.
Use data from **Table 1**.

[2 marks]

Time = _____ min

11

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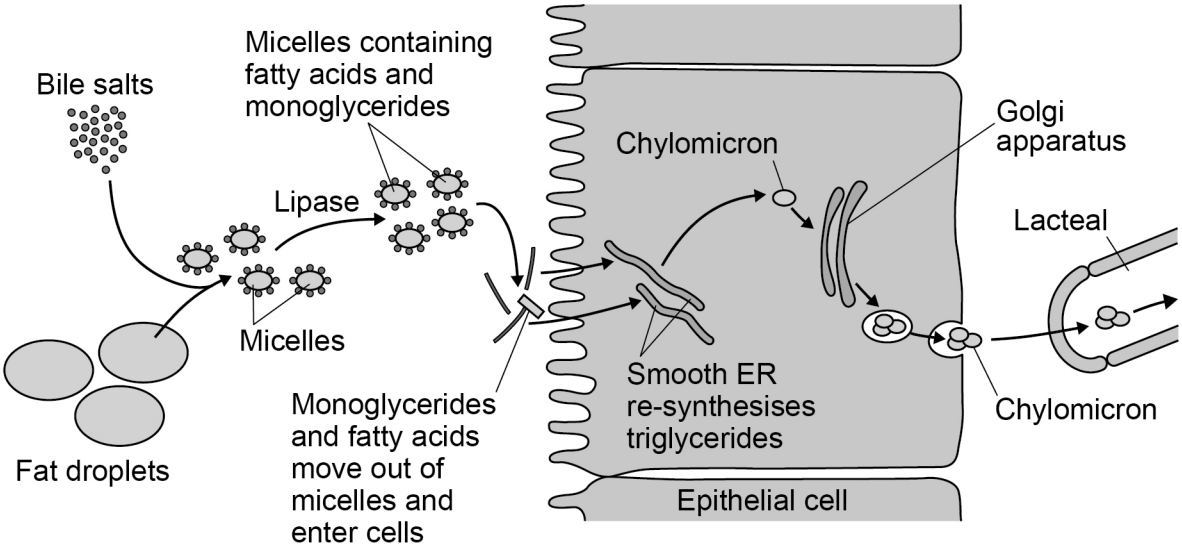
Turn over for the next question

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ANSWER IN THE SPACES PROVIDED

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0 2 Most of the fat in the human diet is in the form of triglycerides.
Figure 2 shows the processes involved in the digestion and absorption of fat.

Figure 2



Key

ER = endoplasmic reticulum

0 2 . 1 Describe how the movement of fatty acids and monoglycerides into the epithelial cell is different from the movement of chylomicrons out of the epithelial cell.

[2 marks]

Do not write outside the box

0 2 . 2 What is the role of bile salts?
Tick (✓) **one** box.

[1 mark]

Condensation ☐

Emulsification ☐

Esterification ☐

Hydrolysis ☐

Question 2 continues on the next page

Do not write outside the box

Some students investigated the effects of lipase and bile salts on triglyceride digestion.

The students:

- placed 5 cm^3 of full-fat milk + 7 cm^3 of sodium carbonate solution into each of five test tubes
- added six drops of phenolphthalein indicator solution to each tube
- placed all five test tubes in a water bath at 40 °C for 10 minutes
- added different substances to each tube, as stated in **Table 2**
- measured the time taken for the colour of the liquid in each tube to change from pink (pH>8.5) to white (pH<8.5).

Table 2 shows the results.

Table 2

Test tube	Contents in addition to 5 cm^3 of full-fat milk + 7 cm^3 of sodium carbonate solution	Time taken to change from pink to white / minutes
1	1 cm^3 lipase solution + 1 cm^3 water	8
2	1 cm^3 bile salts solution + 1 cm^3 water	>20
3	1 cm^3 lipase solution + 1 cm^3 bile salts solution	6
4	1 cm^3 boiled lipase solution + 1 cm^3 bile salts solution	>20
5	1 cm^3 lipase solution + 1 cm^3 boiled bile salts solution	6

0 2 . 3 The students measured the time taken for the colour of the liquid to change from pink to white.

Explain why the students did this.

[3 marks]

Do not write outside the box

0 2 . 4 Give the reason why the students included each of the following pairs of test tubes in the investigation.

[3 marks]

Test tube **1** and test tube **2** _____

_____

_____

Test tube **3** and test tube **4** _____

_____

_____

Test tube **3** and test tube **5** _____

_____

_____

0 2 . 5 The students concluded that bile salts increase the rate of triglyceride digestion by lipase.

Give evidence for this conclusion from **Table 2**.

[1 mark]

_____

_____

_____

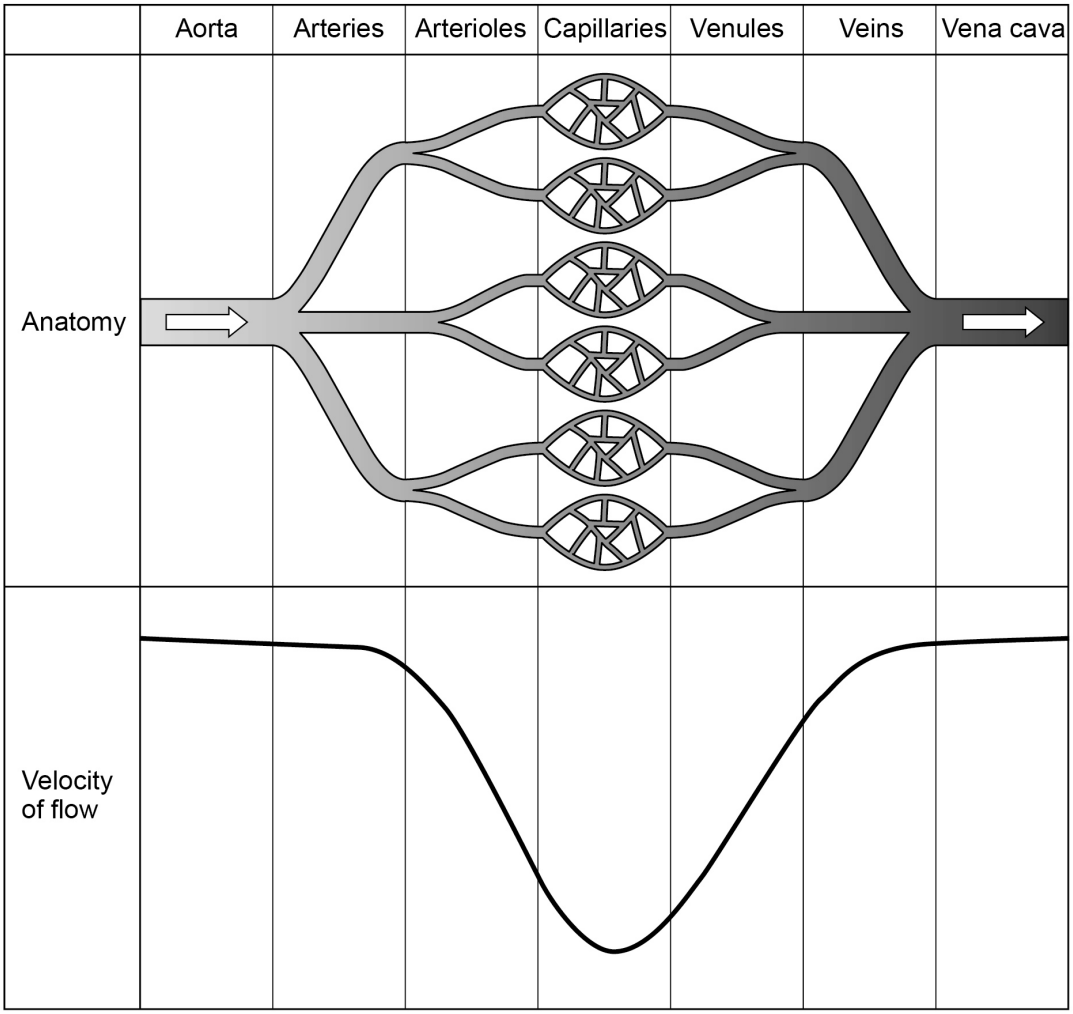
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Turn over for the next question

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0 3 **Figure 3** shows the cross-sectional area and velocity of blood flow in different types of blood vessels.

Figure 3



0 3 . 1 Give **two** differences in the structure of the aorta and the vena cava.

[2 marks]

1 _____

_____

_____

2 _____

_____

Do not write outside the box

0 3 . 2 The velocity of flow decreases as the blood moves from the aorta to the capillaries. Use information from **Figure 3** to explain why.

[2 marks]

0 3 . 3 It is important for blood to flow more slowly in capillaries than in arteries.

Give the reason why.

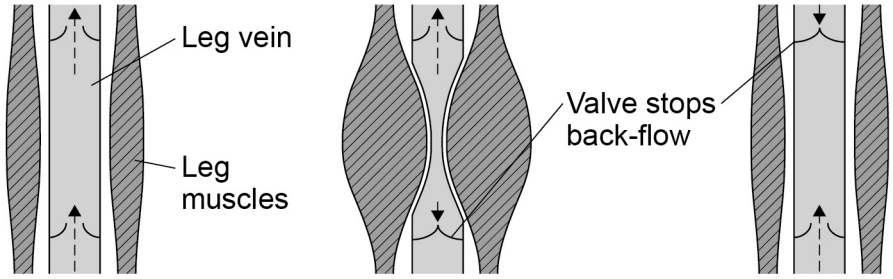
[1 mark]

Question 3 continues on the next page

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0 3 . 4 Sitting on a long plane flight can increase the chance of developing blood clots in the legs. Airlines advise passengers to do leg muscle exercises to reduce the risk of this happening. **Figure 4** shows the effect of leg muscle exercises on a vein in the leg to help blood to return to the heart.

Figure 4



Use information from **Figure 4** to explain how doing these leg muscle exercises helps blood to return to the heart.

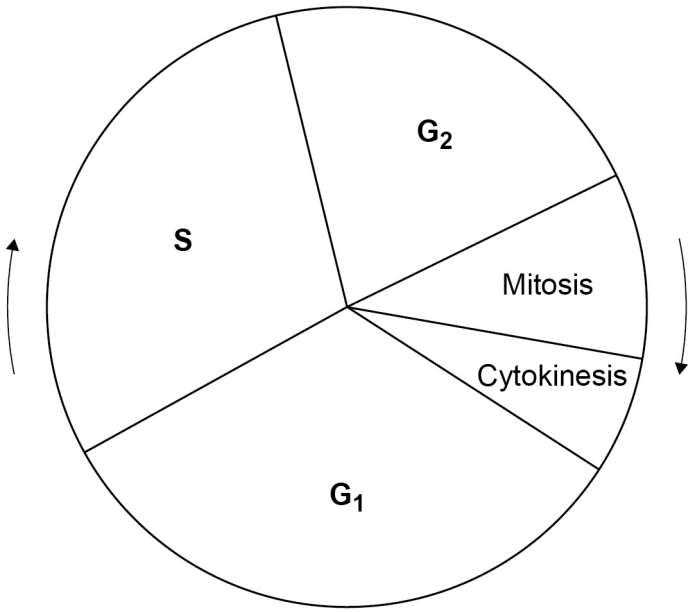
[4 marks]

9

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0 4 **Figure 5** shows the phases of the cell cycle for a slowly dividing eukaryotic cell.

Figure 5



0 4 . 1 Name the phase of the cell cycle that has been sub-divided into G_1, S and G_2.

[1 mark]

_____

0 4 . 2 Describe **two** processes that occur inside the cell during G_1.

[2 marks]

1 _____

_____

2 _____

_____

0 4 . 3 The uncontrolled growth of cells causes a disease called cancer. A drug called Cytarabine disrupts DNA replication, killing the cancer cells.

Name the specific phase of the cell cycle that Cytarabine affects.
Choose your answer from the five phases labelled in **Figure 5**.

[1 mark]

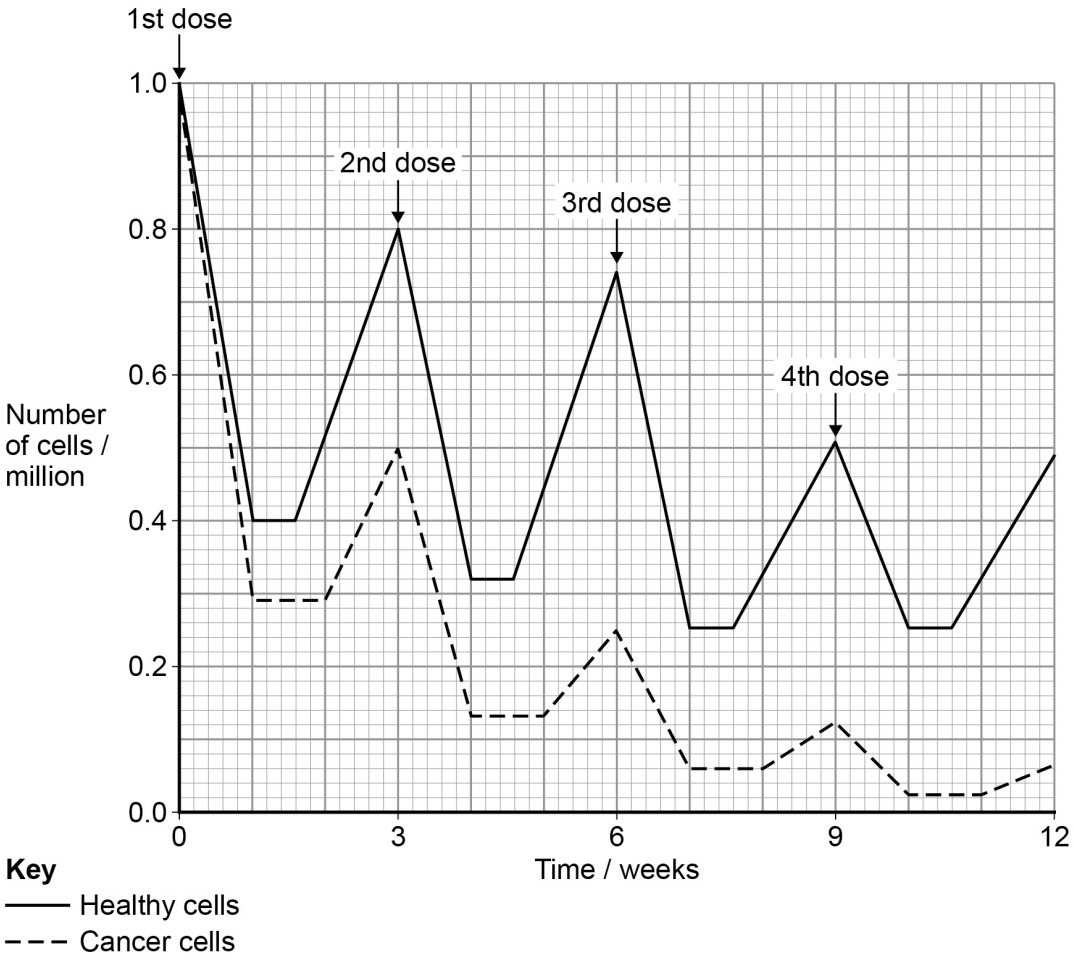
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Question 4 continues on the next page

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Cytarabine was given to a cancer patient once every 3 weeks.
Figure 6 shows the changes in the number of healthy cells and cancer cells over 12 weeks of treatment.

Figure 6



0 4 . 4 Use information from **Figure 6** to calculate the number of cancer cells at week 12 as a percentage of the original number of cancer cells.

[1 mark]

Cancer cells at week 12 = _____ %

Do not write outside the box

0 4 . 5 At the end of week 1, Cytarabine increases the death rate of cancer cells by a factor of 1.18 compared to healthy cells.

By what factor, compared to healthy cells, does Cytarabine increase the death rate of cancer cells at the end of week 12?

Give your answer to three significant figures.

[2 marks]

Factor = _____

0 4 . 6 **Figure 6** shows that Cytarabine has an increasing effect on cancer cells compared to its effect on healthy cells over the 12 weeks of treatment.

Use information from **Figure 6** and your own knowledge to suggest why.

[2 marks]

_____

_____

_____

_____

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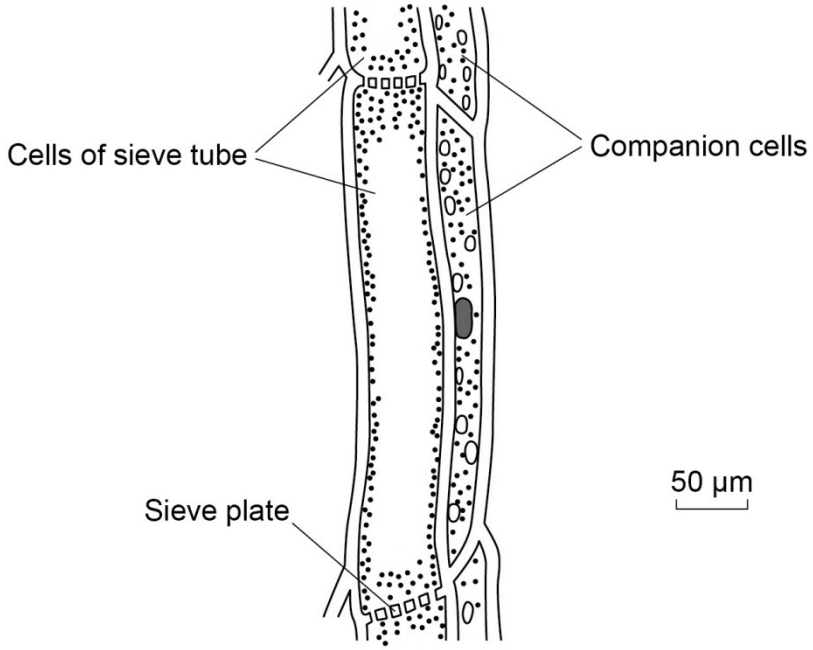
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0 5 In plants, mass transport of organic substances takes place through columns of cells in the phloem that form sieve tubes.

Figure 7 shows some cells from phloem tissue.

Figure 7



0 5 . 1 Sieve plates at the ends of the cells of the sieve tubes allow strands of cytoplasm to connect one cell with the next.

Suggest how this allows mass transport to occur.

[1 mark]

Do not write outside the box

0 5 . 2 Companion cells transport organic substances into and out of the cells of the sieve tubes. Companion cells contain many mitochondria.

Adding a metabolic inhibitor to the phloem prevents respiration.

Explain why metabolic inhibitors reduce the mass transport of organic substances in the cells of the sieve tubes.

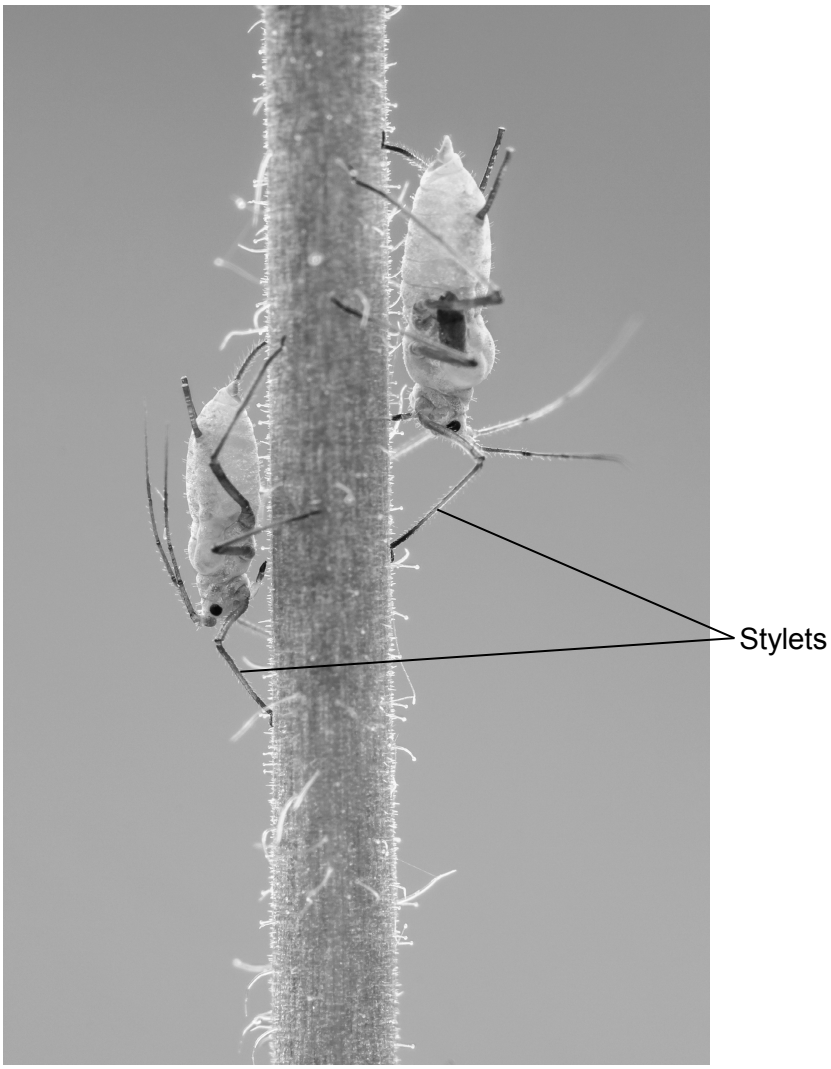
[3 marks]

Question 5 continues on the next page

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Figure 8 shows aphids feeding on the sap of a plant. Aphids have sucking mouthparts called stylets.

Figure 8



A scientist:

- added radioactive sucrose (^{14}C-sucrose) to a leaf **above** the aphid stylet on one plant
- added ^{14}C-sucrose to a leaf **below** the aphid stylet on a second plant
- removed the body of a feeding aphid on each plant, leaving the stylet attached to the plant
- observed that drops of liquid came out of the stylets for several hours
- tested the liquid from both plants and found radioactive sucrose in it.

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0 5 . 3 Complete **Table 3** to suggest **two** conclusions the scientist can make.

[2 marks]

Table 3

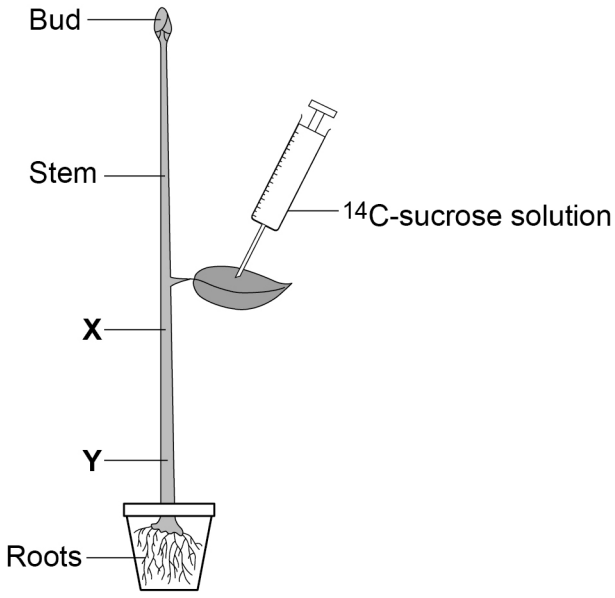
Observation	Conclusion
1. Drops of liquid continued to leave the stylet.	
2. Radioactive sucrose was found in the liquid coming out of the stylets from **both** plants.	

Question 5 continues on the next page

Do not write outside the box

Figure 9 shows a plant leaf supplied with ^{14}C-sucrose.

Figure 9



The scientist wants to determine the rate of transport in the phloem. He takes samples of sap from aphid stylets at two points, **X** and **Y**, as shown in **Figure 9**. The scientist analyses the sap for radioactive substances.

0 5 . 4 Give **two** other measurements the scientist would need to make to determine the rate of transport in the phloem.

[2 marks]

1 _____

_____

2 _____

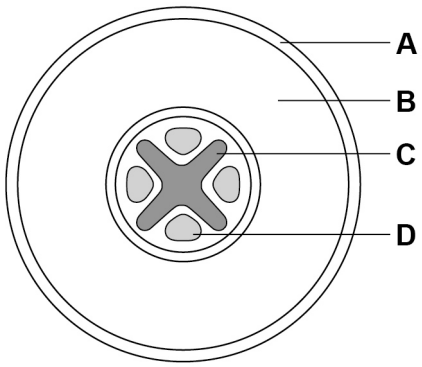
_____

Do not write outside the box

After 20 minutes, the scientist removed a thin section of the plant's root and identified the areas containing ^{14}C-sucrose.

Figure 10 shows a cross-section through a dicotyledonous root.

Figure 10



0 5 . 5 Give the letter, **A**, **B**, **C** or **D**, of the part of the root that would contain the highest level of radioactivity.

[1 mark]

9

Turn over for the next question

Do not write outside the box

0 6 A student investigated the effect of exercise on heart rate.

The student:

1. sat down and rested for 10 minutes
2. measured the resting pulse rate at the wrist for 20 seconds
3. repeated step 2 four more times, at 1-minute intervals
4. did some exercise
5. immediately after the exercise, measured the pulse rate for 20 seconds.

0 6 . 1 Why was it important to measure the resting pulse rate?

[1 mark]

0 6 . 2 Give **two** ways of standardising the exercise so another student could repeat the investigation in exactly the same way.

[2 marks]

1

2

The student measured the pulse rate for 20 seconds at the end of exercise and then calculated the heart rate in beats per minute.

0 6 . 3 Suggest why measuring pulse rate for less than 20 seconds could give inaccurate results for heart rate.

[1 mark]

Do not write outside the box

Table 4 shows the student's pulse rate at rest and immediately after exercise. It also shows stroke volume and cardiac output.

Table 4

	Pulse rate / beats min^{-1}	Stroke volume / cm^3 $beat^{-1}$	Cardiac output / cm^3 min^{-1}
At rest	75	75	5 625
Immediately after exercise	180		21 600

0 6 . 4 Calculate the stroke volume of the student immediately after exercise.

[1 mark]

Stroke volume = _____ cm^3 $beat^{-1}$

0 6 . 5 Calculate the percentage increase in the student's cardiac output from at rest to immediately after exercise.

[1 mark]

Percentage increase = _____ %

Question 6 continues on the next page

Do not write outside the box

0 6 . 6 Suggest the advantage of an increased cardiac output to muscles **during** exercise.

[3 marks]

9

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0 7 Ebola is a serious viral illness. There was a large outbreak of Ebola in 2014 in West Africa that killed many people.

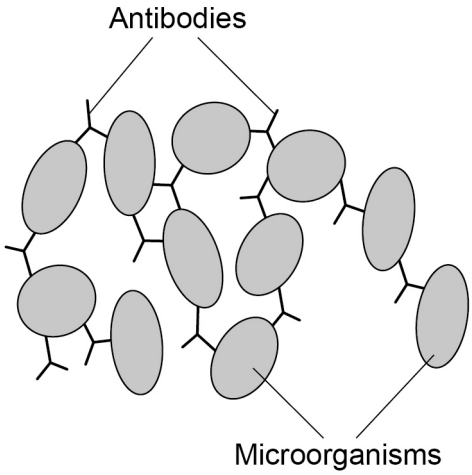
Vaccines and drug therapies are being developed and tested against the protein antigens on the surface of the Ebola virus.

0 7 . 1 What is an antigen?

[1 mark]

Figure 11 shows how antibodies can stick microorganisms together in large clumps.

Figure 11



0 7 . 2 Name the process shown in **Figure 11**.

[1 mark]

0 7 . 3 How does the process shown in **Figure 11** help in the destruction of the microorganisms?

[1 mark]

Do not write outside the box

0 7 . 4 Antiserum is blood plasma containing antibodies. One treatment for Ebola infection is to inject the antiserum from a recovering Ebola patient into someone who has the infection.

Suggest **two** reasons why injecting antiserum is **not** a very successful treatment for Ebola.

[2 marks]

1 _____

_____

_____

2 _____

_____

_____

0 7 . 5 Scientists are developing a vaccine for Ebola.

Explain how vaccination would control the spread of Ebola.

[3 marks]

_____

_____

_____

_____

_____

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_____

8

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0 8 Water enters a plant from the soil. The water travels up through the plant and eventually evaporates into the air.

0 8 . 1 Describe the pathway that water takes from the soil to the endodermis.

[4 marks]

Question 8 continues on the next page

Do not write outside the box

0 8 . 2 Describe and explain the effect of each of the following environmental factors on the rate of transpiration:

- increased air temperature
- increased light intensity
- increased humidity of the air.

[6 marks]

END OF QUESTIONS

10

Figure 1: Getty Images/Ed Reschike
Figure 8: Aphids © Heath McDonald/Science Photo Library

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